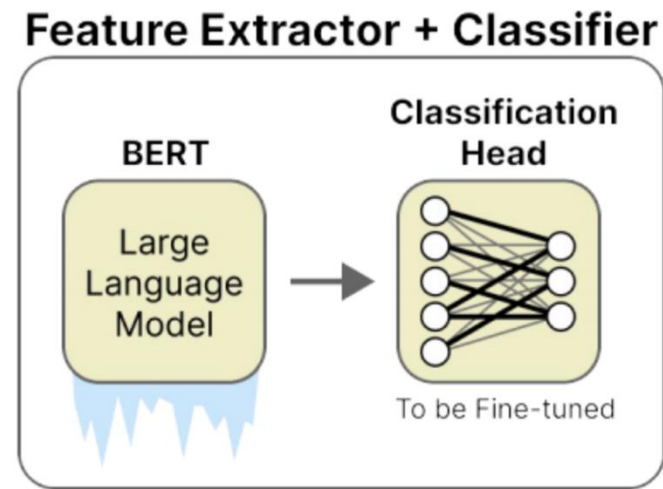


Globally Complete, Locally Grounded: SHAP-Guided Concept Discovery

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Deep neural networks achieve outstanding performance but often lack transparency. This presentation introduces a novel approach that integrates local attributions from SHAP into concept discovery for neural networks, creating explanations are both globally sufficient and locally grounded.

Our method identifies influential segments using SHAP values, clusters them to obtain meaningful centers, and then aligns concept vectors with these centers. The result is a set of interpretable concepts that maintain high completeness while anchored to important model features.

Related Work

Feature-Based Explanation Methods

Techniques: LIME, SHAP

Key Point: Assign importance scores to input features, offering local explanations but lacking insight into the global reasoning process.

Counterfactual Explanation Methods

Focus: Determine "what to change" in the input to alter the model's output

Key Point: Provide actionable insights yet fall short in revealing the underlying factors driving predictions.

Sample-Based Explanation Methods

Technique: Explaining predictions by identifying similar training examples

Key Point: Intuitive but do not provide a formal global framework.

Concept-Based Explanations

Techniques:

- TCAV uses human-annotated concepts for directional sensitivities.
- ACE automatically discovers concepts via clustering image patches.
- Unsupervised methods (e.g., PCA) extract latent factors, though they may not align with human semantics.

Key Point: Unsupervised semantic discovery often requires inductive biases to achieve meaningful results.

The Challenge of Neural Network Interpretability



Black Box Nature

Deep neural networks achieve state-of-the-art performance across domains but their complex internal workings remain opaque to humans.



Local vs. Global Explanations

Traditional methods like SHAP provide local explanations for individual predictions but often fail to capture the global reasoning of the model.



Human-Friendly Concepts

Humans prefer to explain decisions using high-level concepts—grouping similar examples and reasoning in terms of abstract ideas.

Concept-Based Explanations

Concept-based explanations provide human-friendly insights by discovering vectors that represent high-level factors in the latent space. These approaches assume that concepts lie in a low-dimensional subspace of an intermediate DNN layer.



Completeness

How sufficient concepts are for reconstructing model decisions



Interpretability

How meaningful concepts are to humans



Representation

How concepts capture latent space directions

Our Approach: Bridging Local and Global Explanations

1. Fine-tuning & Activation Extraction

- Fine-tune a BERT model for sentiment classification.
- Extract intermediate activations to serve as feature representations.

2. SHAP Value Computation & Clustering

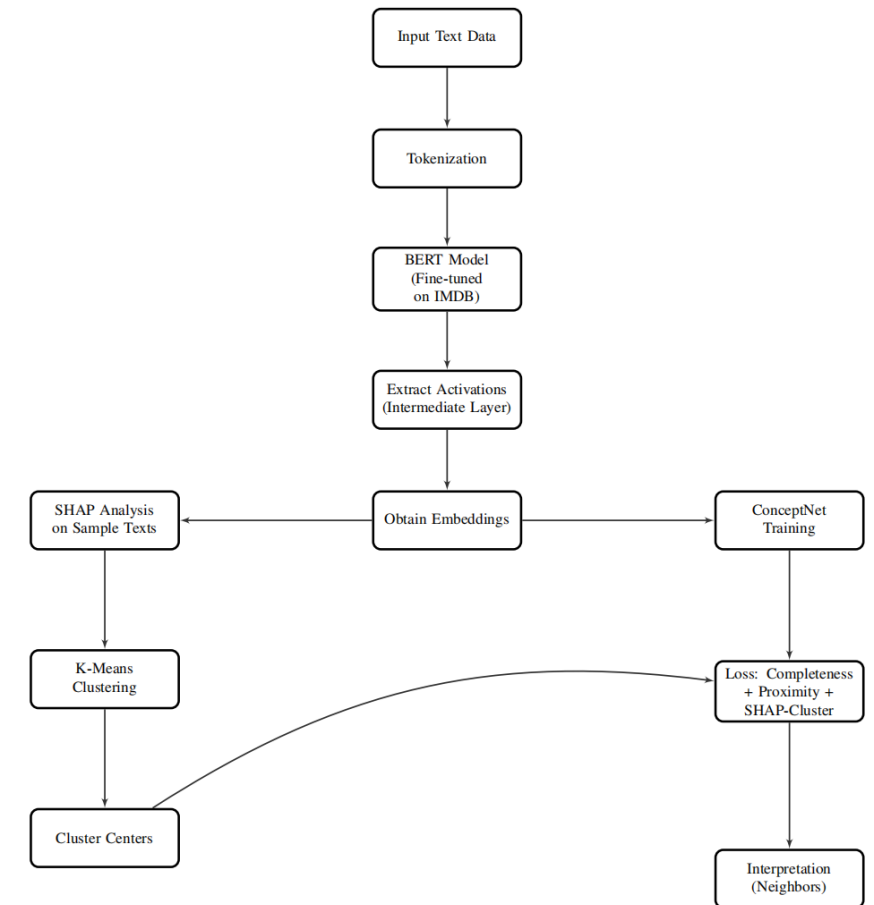
- Compute sentence-level SHAP values to identify influential features.
- Cluster the high-level embeddings to derive semantic cluster centers that capture key patterns.

3. ConceptNet Model Training with Composite Loss

- Train a ConceptNet model using a composite loss function that includes:
- Proximity Regularizer: Ensures concept vectors are close to the corresponding high-level features.
- SHAP-Cluster Regularizer: Enforces alignment between the cluster centers and the learned concepts.

4. Concept Interpretation via Nearest Neighbor Analysis

- Analyze the learned concept vectors by finding their nearest neighbors to interpret and validate the semantic meaning of each concept.



BERT Fine-Tuning & Activation Extraction

Pre-trained Model & Task

- Use a pre-trained BERT model fine-tuned on IMDB sentiment classification.
- Each review x_i is tokenized; sentiment label $y_i \in \{0,1\}$.

Model Structure

- Defined as:

$$f(x) = h(\phi(x)),$$

where $\phi(x) \in \mathbb{R}^d$ is an intermediate embedding and $h(\cdot)$ is the classification head.

Training Objective

- Minimize standard cross-entropy loss

$$\mathcal{L}_{\text{BERT}} = -\frac{1}{n} \sum_{i=1}^n \log P(y_i \mid x_i).$$

Activation Extraction

- Use a forward hook to extract activations from the penultimate layer for later use.

SHAP Analysis & Cluster Center Extraction

SHAP Value Computation

- Compute sentence-level SHAP values for a subset of training samples.
- For a token i in sentence x , the SHAP value is:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} \left(f(S \cup \{i\}) - f(S) \right),$$

Importance Score & Selection

- Aggregate absolute SHAP contributions for the positive class each sentence:

$$I(x) = \sum_{i=1}^{|x|} |\phi_i^{(1)}|.$$

- Rank sentences by $I(x)$ and select the top-k

Clustering for Semantic Centers

- Cluster the corresponding BERT embeddings using K-means.
- The resulting cluster centers $\mu_1, \mu_2, \dots, \mu_N$ capture dominant high-level semantics.

ConceptNet Training Overview

Objective

- Learn a set of m concept vectors $\{c_1, c_2, \dots, c_m\} \subset \mathbb{R}^d$ such that the concept score
 - $v_c(x) = (\langle \phi(x), c_1 \rangle, \langle \phi(x), c_2 \rangle, \dots, \langle \phi(x), c_m \rangle)$ is a sufficient statistic for predicting y.

Mapping to Prediction

- Assume existence of a mapping $g: \mathbb{R}^m \rightarrow \mathbb{R}^d$ so that:

$$h(g(v_c(x))) \approx f(x), \quad \forall x \in \mathcal{D}.$$

Overall Objective Function

- Maximize the log-likelihood of predictions with the composite regularizer:

$$\max_{c_1, \dots, c_m, g} \log P(y \mid g(v_c(x))) + R_{\text{reg}},$$

$$R_{\text{reg}} = R_{\text{prox}} - R_{\text{shap}}.$$

Proximity Regularization

Purpose

- Encourages each concept vector to be close to its top-K nearest neighbors (patches or segments) in the training data.
- Simultaneously maintains diversity among different concept vectors.

Mathematical Formulation

- For each concept c_k denote T_{c_k} its set of top-K nearest neighbors.
- The proximity regularizer is defined as:

$$R_{\text{prox}} = \lambda_1 \left(\frac{1}{m} \sum_{k=1}^m \frac{1}{K} \sum_{x \in T_{c_k}} \langle \phi(x), c_k \rangle \right) - \lambda_2 \left(\frac{1}{m(m-1)} \sum_{j \neq k} \langle c_j, c_k \rangle \right),$$

λ_1 and λ_2 are hyperparameters that balance the two terms.

SHAP-Cluster Regularization & Final Loss

SHAP-Cluster Regularization

- Aligns learned concept vectors with the cluster centers from the SHAP analysis.
- The SHAP-cluster loss is:

$$R_{\text{shap}} = \lambda_{\text{shap}} \cdot \frac{1}{N} \sum_{i=1}^N \min_{j \in \{1, \dots, m\}} \|c_j - \mu_i\|_2,$$

This term minimizes the average distance between each cluster center and its closest concept vector.

Final Objective

- The complete loss function is:

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_1 R_{\text{prox}} - \lambda_{\text{shap}} R_{\text{shap}},$$

Where:

- $\mathcal{L}_{\text{pred}}$ is the prediction loss (cross-entropy).
- The composite loss enforces that the concept scores are both predictive and semantically aligned with important features.

Interpretation and Analysis of Concept Vectors

•Concept Loading:

- Load learned concept vectors from a NumPy file (shape: $m \times d$, where m = number of concepts, d = embedding dimension).

•Embedding Preparation:

- Convert training embeddings (BERT activations) into a torch tensor for analysis.

•Nearest Neighbor Search:

- Compute Euclidean distances between each concept vector and all training embeddings.
- Retrieve the top 150 training examples (neighbors) with the smallest distances for each concept.

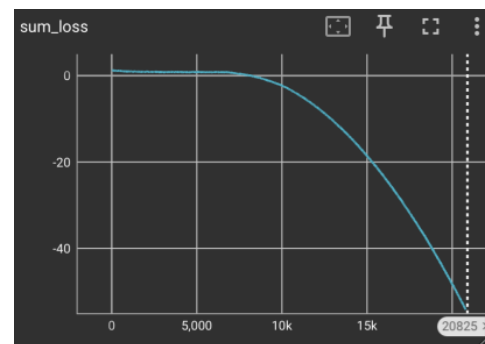
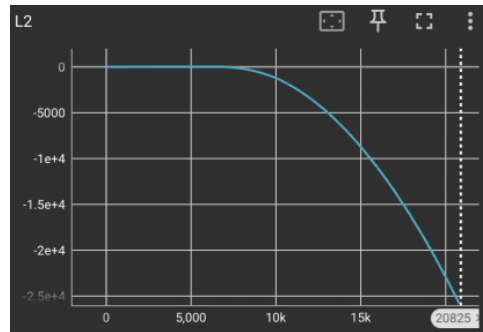
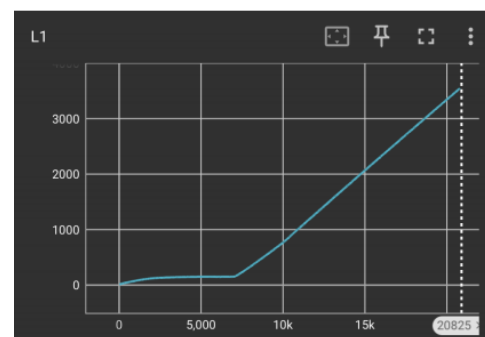
•Word Extraction & Frequency Analysis:

- Extract words from the input sentences corresponding to the top-150 neighbors.
- Filter out common stop words and punctuation, then count word frequencies.

•Concept Characterization:

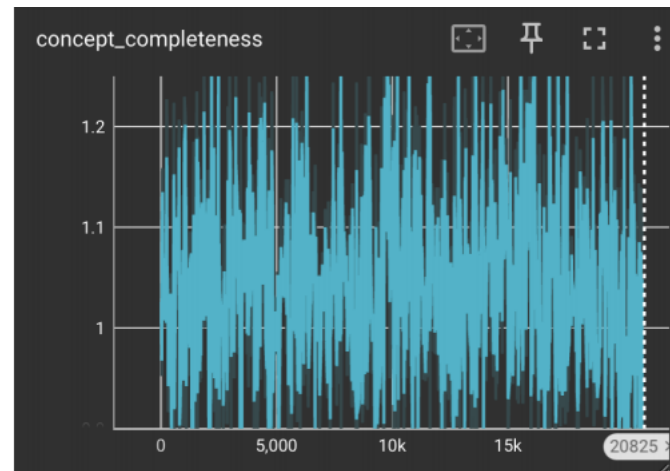
- Report the 25 most common words for each concept to interpret its semantic meaning.

Evaluation Results



Training Metrics

TensorBoard logs show the proximity term (L1) increasing as concept vectors move closer to their neighbors, while the orthogonality term (L2) decreases as concepts become more distinct from each other, and the total loss with shap term.



Completeness Score

The completeness metric stabilizes around a value close to 1.0, indicating that our concept-based projection retains most of the predictive power of the original BERT model.

$$\text{Completeness}(y_{\text{pred}}) = \frac{\text{new_correct} - \frac{1}{m}}{\text{orig_correct} - \frac{1}{m}},$$

- orig_correct is the number of correct predictions by the original BERT model on that batch,
- new_correct is the number of correct predictions using only the concept-based projection,

Concept	Top Words (word, frequency)
Concept 1	dull (27), really (24), stupid (24), movie (22), pointless (19), plot (12), just (11), />br (9), it. (8), thing (8), annoying (8), terrible (7), calculated (6), imitation (6), disguised (6), poorly (6), acted (6), clearly (6), biased (6), funny (6), got (6), idea (6), weak (6), unfortunately (6), it's (6)
Concept 2	dull (23), just (22), really (22), stupid (19), acting (13), horrible (13), movie (13), poorly (11), bad (10), funny (9), absolutely (9), boring (9), ridiculous (8), plot (8), all. (7), story (7), movie. (7), />br (7), simply (6), lacks (6), predictable (6), terrible (6), big (6), disappointment (6), annoying (6)
Concept 3	story (9), good (8), movie (7), better (7), just (6), pretty (6), interesting (5), acting (5), performances (5), film (5), great (4), real (4), quite (4), />br (4), truly (3), character (3), script (3), way (3), hilarious (3), thing (3), act (2), musical (2), strange (2), make (2), world (2)
Concept 4	love (26), wonderful (21), real (18), story (15), really (13), life (13), truly (11), movie (10), entertaining (9), wild (7), good (7), leaves (7), strange (6), weird (6), character (6), moved (5), brave (5), little (5), boy (5), exceptional (5), beautiful (5), gives (5), man (5), book (5), feeling (5)
Concept 5	love (24), wonderful (22), real (18), story (16), really (13), life (13), truly (12), movie (10), entertaining (9), wild (7), good (7), strange (6), weird (6), character (6), man (6), leaves (6), great (6), exceptional (5), moved (5), brave (5), little (5), boy (5), gives (5), beautiful (5), way (5)

Concept Analysis

Semantic analysis of top neighbor patches confirms that each concept captures a coherent theme, ranging from strongly negative sentiment ("dull", "stupid") to positive sentiment ("love", "wonderful").

Conclusions and Future Directions

5

Discovered Concepts

Interpretable concept vectors that strongly correlate with sentiment

~1.0

Completeness Score

Indicating minimal loss of predictive power

Key Innovations

Local proximity, orthogonality with SHAP-based regularization

We successfully integrated local attributions from SHAP into concept discovery, creating explanations that are both globally sufficient and locally grounded. Our method bridges concept-based interpretability with state-of-the-art language models, ensuring that learned concepts remain both semantically meaningful and highly predictive.

Future directions include extending to multi-task or multi-domain settings, incorporating concept-based interpretability into generative language models, and integrating concept regularization directly into model fine-tuning for more robust or disentangled concepts.

Thank you

https://github.com/lucasli05/XAI_Shap-concept_NN_for_LLM